M. Ikram Ullah

ML Researcher · Multimodal Generation, LLM Evaluation & Applied Deep Learning

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SUMMARY

Machine-learning researcher (MS Data Science, Quaid-e-Azam University) working on low-resource multimodal generation and on measuring LLM systems rigorously. My MS thesis turns complete Urdu stories into right-to-left comics with open-weight models, with paired-bootstrap comparisons against a baseline. Earlier work compares CNNs and Vision Transformers on imbalanced facial-emotion data; applied work includes explainable chest X-ray classification (Grad-CAM) and summarization and topic-model comparisons. I design evaluations with held-out splits, ablations, confidence intervals and validated LLM judges.

RESEARCH

UrduComicLab: complete-story Urdu story-to-comic generation (MS thesis, in progress)

- Turns complete Urdu children's stories into right-to-left comic books with correctly shaped Nastaliq dialogue, narration captions and sound effects, using open-weight models only (Gemma-2-9B parser, SDXL with IP-Adapter, Grounding DINO bubble placement, HarfBuzz lettering).
- Against a baseline on 15 paired comics (3 stories × 5 seeds, paired bootstrap CIs): spoken lines printed 0.49 → 1.00, dialogue F1 0.63 → 1.00, speaker accuracy 0.83 → 1.00, face consistency (DINOv2) 0.54 → 0.67; about 35 pinned open models benchmarked across four experiment campaigns.

Facial emotion recognition on imbalanced FER2013: CNN vs Vision Transformer (paper, pending publication)

- Compared an enhanced ResEmoteNet CNN with squeeze-and-excitation blocks against ViT-Base/16 on FER2013 (35,887 images, 7 classes, 16.5:1 class ratio), with class-balanced focal loss, two-stage training and CutMix, Mixup and RandAugment.
- Showed that ViT self-attention avoids minority-class collapse (64.9% vs 0% accuracy on Disgust), while the CNN with channel attention was stronger overall on low-resolution faces.

EDUCATION

MS Data Science, Quaid-e-Azam University, Islamabad — Feb 2025 – Present

Coursework: Machine Learning, Deep Learning, NLP, Big Data Analytics, Statistical Modeling, Data Visualization. Thesis: UrduComicLab.

PROJECTS

Data Analyst Agent (text-to-SQL) *(github.com/ikram4748/data-analyst-agent)*

LangGraph · Gemini · sqlglot · DuckDB · fastembed · Streamlit

- Built a self-correcting text-to-SQL agent: local-embedding schema linking, a two-layer read-only SQL guard, a literal-value check that catches silent zero-result errors, and a bounded fix loop driven by database feedback.
- Raised execution accuracy on 150 held-out BIRD questions from 60.7% (one-shot baseline, same model) to 66.7% (+6.0 points, 95% CI +1.3 to +11.3) at 1.09 LLM calls per question; tuned on a separate dev split after error analysis of 48 failures.
- Found and fixed a measurement bug in which rate-limit fallback silently mixed three models across benchmark variants.

Literature Review Agents *(github.com/ikram4748/research-assistant-agent)*

LangGraph · Gemini · arXiv & OpenAlex APIs · fastembed

- Built a multi-agent reviewer (planner, searcher, reader, writer, critic, reviser) in which every finding must quote its source abstract word for word, checked in code, and a critic sends unsupported sentences back for revision.
- Evaluated citation grounding against a single-agent baseline with an independent LLM judge whose sensitivity is validated by swapping citations to wrong papers.
- On 20 held-out questions both systems reached 97% sentence support; multi-agent cited more precisely (87% vs 80%), but the paired 95% CI included zero at 6× the LLM calls. Fixed a judge that passed every sentence: its catch rate for deliberately wrong citations rose from 48% to 90%.

Financial RAG Lab *(github.com/ikram4748/financial-rag-lab)*

PyMuPDF · BM25 · bge-small · cross-encoders · Gemini

- Built a RAG benchmark on FinanceBench (150 analyst questions, 84 filings, 12,013 pages) comparing 3 chunking strategies, 3 metadata scopes and 5 retrievers (BM25, dense, hybrid RRF, two cross-encoder rerankers) plus HyDE, scored on evidence-page recall with a dev/test split.

- On 100 held-out questions, dense retrieval with company/year filtering beat BM25, hybrid and cross-encoder reranking; HyDE raised evidence recall@5 from 44% to 60% and answer accuracy from 47% to 61% (+14 points, 95% CI +5 to +23), vs 24% for naive RAG and 84% with the evidence pages given, which showed retrieval was the bottleneck. Answers graded by a separate judge model validated on gold and corrupted answers.

E-mail Experiment & Uplift Modeling *(github.com/ikram4748/uplift-ab-testing)*
Python · statsmodels · LightGBM · Plotly · Streamlit

- Analysed a 64,000-customer randomized e-mail experiment end to end: sample-ratio and balance checks, Holm-corrected effects (men's e-mail +7.7 points visits, +$0.77 spend per customer, 95% CI $0.49–$1.05), CUPED and regression adjustment, power analysis, and a peeking simulation (10 looks → 18% false positives).
- Built S/T/X-learner and transformed-outcome uplift models with Qini evaluation over 5 splits and valued targeting policies by inverse-propensity weighting; showed that e-mailing everyone beats model targeting below about $0.23 per e-mail.

CLIENT PROJECTS (CODE PRIVATE)

- **MedVision (chest X-ray classification)** *(TensorFlow/Keras · ResNet50 · EfficientNetB0 · Grad-CAM)*: Pneumonia detection on 5,856 clinical X-rays comparing a CNN from scratch, frozen ResNet50 and fine-tuned EfficientNetB0; Grad-CAM heatmaps show which lung regions drive each prediction.
- **Stroke-Risk Prediction** *(scikit-learn · XGBoost · imbalanced-learn · SHAP)*: Random Forest vs XGBoost on patient data with a 5% positive class; SMOTE on the training split only (no leakage), decision threshold tuned on the precision-recall curve, SHAP explanations per patient.
- **TextMiner (summarization and topics)** *(T5 · TextRank · LDA · BERTopic)*: Extractive (TextRank, LSA) vs abstractive (T5) summarization scored with ROUGE, and LDA vs BERTopic topic models checked against true categories (ARI, NMI) on 300+ news articles.

EXPERIENCE

AI/ML Engineer, Syslab Technologies *(Islamabad, Pakistan)* Oct 2024 – Nov 2025

- Architected VSARA, an agentic healthcare voice and chat assistant combining a RAG pipeline (FAISS + Hugging Face sentence-transformers) with Whisper speech recognition and neural text-to-speech.
- Improved diagnostic response accuracy by 23% through systematic A/B testing of model configurations, prompt strategies and RAG retrieval settings (chunk size, retrieval depth).
- Investigated model failures through error logs, edge cases and interaction data; targeted retraining and data-quality fixes reduced the error rate by 18%.

Freelance AI/ML Engineer, Self-employed (direct clients) *(Remote)* Feb 2023 – Present

- Delivered 15+ end-to-end AI/ML projects for healthcare, e-commerce and fintech clients, from scoping and data collection to deployment and handoff.
- Fine-tuned Hugging Face models (BERT, DistilBERT, Mistral-7B) for text classification, sentiment analysis and named-entity recognition; deployed solutions with FastAPI and Docker.

TECHNICAL SKILLS

Research methods: experimental design, ablations, held-out dev/test protocols, paired bootstrap CIs, error analysis, LLM-as-judge validation, reproducible pipelines (pinned model revisions, fixed seeds)
Deep learning: PyTorch, Vision Transformers, CNNs (SE blocks), focal and class-balanced losses, CutMix/Mixup, transfer learning, Grad-CAM, diffusion models (SDXL, FLUX), IP-Adapter, Grounding DINO, DINOv2
NLP & LLMs: open-weight LLMs (Gemma, Qwen, Llama), Urdu/Nastaliq NLP and RTL text shaping (HarfBuzz), RAG, LLM agents (LangGraph), Hugging Face Transformers, T5, LDA/BERTopic, ROUGE
Statistics & data: causal inference, A/B testing, uplift modeling, Python, SQL, pandas, statsmodels, scikit-learn, LightGBM
Engineering: GPU experiment campaigns, Git, Docker, FastAPI, pytest

CERTIFICATIONS

Machine Learning Specialization · Deep Learning Specialization · Natural Language Processing · Big Data Analytics